Evaluate yourself:

Compare the stability of Ni^{4+} and Pt^{4+} from their ionisation enthalpy values.

IE	Ni	Pt
I	737	864
II	1753	1791
III	3395	2800
IV	5297	4150

4.3.4 Oxidation state:

The first transition metal Scandium exhibits only +3 oxidation state, but all other transition elements exhibit variable oxidation states by losing electrons from (n-1)d orbital and ns orbital as the energy difference between them is very small. Let us consider the 3d series; the following table summarizes the oxidation states of the 3d series elements.

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
				+7				
			+6	+6	+6			
		+5	+5	+5				
	+4	+4	+4	+4	+4	+4	+4	
+3	+3	+3	+3	+3	+3	+3	+3	
	+2	+2	+2	+2	+2	+2	+2	+2
								+1

At the beginning of the series, +3 oxidation state is stable but towards the end +2 oxidation state becomes stable.

The number of oxidation states increases with the number of electrons available, and it decreases as the number of paired electrons increases. Hence, the first and last elements show less number of oxidation states and the middle elements with more number of oxidation states. For example, the first element Sc has only one oxidation state +3; the middle element Mn has six different oxidation states from +2 to +7. The last element Cu shows +1 and +2 oxidation states only.

The relative stability of different oxidation states of 3d metals is correlated with the extra stability of half filled and fully filled electronic configurations. Example: $Mn^{2+}\left(3d^5\right)$ is more stable than $Mn^{4+}\left(3d^3\right)$

The oxidation states of 4d and 5d metals vary from +3 for Y and La to +8 for Ru and Os. The highest oxidation state of 4d and 5d elements are found in their compounds with the higher electronegative elements like O, F and Cl. for example: RuO_4, OsO_4 and WCl_6. Generally in going down a group, a stability of the higher oxidation state increases while that of lower oxidation state decreases.It is evident from the Frost diagram (ΔG^0 vs oxidation number) as shown below,For titanium,vanadium and chromium, the most thermodynamically stable oxidation state is +3. For iron, the stabilities of +3 and +2 oxidation states are similar.Copper

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